

Governing the AI Shock: From Laissez-Faire Disruption to a Broad-Based AI Social Compact

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Abstract

AI-driven productivity gains need not translate into broadly shared welfare when labor displacement, household demand, firm survival, and public finance adjust jointly. We develop a monthly stock–flow-aware heterogeneous-agent simulation of a stylized urban economy with 500 household decision units, approximately 30 firms, a government, and a deterministic banking sector. Bounded large language model (LLM) decisions affect only pre-specified categorical stances; deterministic code settles all economic quantities and accounts. Following a 24-month common pre-equilibrium, seven institutional counterfactuals run through month 120 under two LLM-participation regimes. In the retained seed, laissez-faire private AI raises cumulative real consumption by 5.4% relative to the no-new-AI path but increases peak unemployment from 2.8% to 13.0%. Employment responsibility blocks 297 AI-attributable layoffs but shifts job losses toward firm exit. An earmarked AI infrastructure levy raises the tail public-service index by 60.4% with little aggregate-demand loss, whereas 75.6% of solo-enterprise sales displace incumbent demand. The integrated compact improves household liquidity and public services but weakens firm survival. These counterfactuals identify mechanism-level trade-offs and testable institutional hypotheses rather than a city forecast or a robust policy ranking.

Keywords: artificial intelligence, technological unemployment, agent-based modeling, stylized urban economy, AI governance

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1. Introduction

1.1. *Urban-science motivation and research questions*

Artificial intelligence is expanding from specialized prediction systems into increasingly versatile technologies that can generate content, assist complex cognitive work, and interact with the physical world. GPT-family large language models have advanced text generation, dialogue, and interactive problem solving (OpenAI, 2023); text-to-image models have extended generative capabilities to visual content (Ramesh et al., 2022); and code-generation systems can now address increasingly complex programming tasks (Li et al., 2022). In parallel, vision-language planning and embodied multimodal models are extending AI into autonomous driving, robotics, and other physical systems (Pan et al., 2024; Driess et al., 2023). Together, these developments point toward a broader technological platform with the potential to reshape production, labor, mobility, public services, and everyday urban life.

As these capabilities diffuse across workplaces and urban systems, however, improvements in task performance do not automatically translate into economy-wide welfare. Generative AI has made the distinction between task productivity and economy-wide welfare unusually salient. Workplace studies show that generative tools can reduce completion time and improve output (Noy and Zhang, 2023; Brynjolfsson et al., 2025), while occupational exposure measures suggest that a large share of work activities may be affected by large language models (LLMs) (Eloundou et al., 2024). These findings leave a central macroeconomic question unresolved: who can sustain demand for the additional output when labor demand, wage income, household liquidity, firm survival, and public finance adjust simultaneously?

Technical change can shift production possibilities (Solow, 1957), but effective demand can constrain employment even when productive capacity is abundant (Keynes, 1936). Consumption responds to expected income, precautionary motives, and liquidity (Friedman, 1957; Deaton, 1991; Carroll, 1997). Task-based approaches further separate displacement, productivity, and reinstatement effects (Autor et al., 2003; Acemoglu and Restrepo, 2018). The distribution of AI gains therefore depends not only on technical capability, but also on institutional rules governing employment adjustment, taxation, public return, entrepreneurship, and fiscal space.

35 This paper addresses four questions. First, what is the joint effect of
 36 private AI on consumption and transition unemployment when no dedicated
 37 social-return policy is added? Second, can employment responsibility, an
 38 AI infrastructure levy, or solo-enterprise formation redirect the productivity
 39 dividend? Third, how do these mechanisms operate as an integrated so-
 40 cial compact, and what changes when fiscal limits tighten? Fourth, are the
 41 resulting paths sensitive to whether firms use bounded LLM decisions?

42 The contribution is fourfold. We construct a closed feedback loop linking
 43 productivity, employment, income, demand, and firm survival rather than
 44 imposing unemployment externally. We evaluate consumption, employment,
 45 household liquidity, distribution, public services, firm dynamics, and mar-
 46 ket structure jointly. We separate deterministic economic settlement from
 47 bounded categorical LLM decisions. Finally, we link substantive claims to
 48 explicit accounting, boundary, pre-equilibrium, decision-fallback, and behav-
 49 ioral qualification evidence.

50 *1.2. Theoretical framework and related literature*

51 *1.2.1. Effective demand and household liquidity*

52 Firms hire and produce in response to expected sales. When labor dis-
 53 placement reduces household income faster than prices or new tasks restore
 54 purchasing power, productive capacity can coexist with insufficient effective
 55 demand (Keynes, 1936). Government procurement enters firm orders di-
 56 rectly, whereas transfers first enter household disposable income and then
 57 consumption. Fiscal multipliers are therefore generated by the model’s feed-
 58 back loop rather than imposed as fixed coefficients.

59 Households distinguish current resources from baseline income and con-
 60 sumption needs, reflecting permanent-income and consumption-smoothing
 61 arguments (Friedman, 1957). Cash buffers, unemployment duration, a min-
 62 imum consumption rule, and precautionary contraction approximate buffer-
 63 stock saving and smoothing under liquidity constraints (Deaton, 1991; Car-
 64 roll, 1997). These rules are heuristic rather than the solution to a fully
 65 specified intertemporal optimization problem.

66 *1.2.2. Technology, tasks, and firm dynamics*

67 AI changes productivity and productive capacity (Solow, 1957), reallo-
 68 cates task content and labor requirements (Autor et al., 2003; Acemoglu and
 69 Restrepo, 2018), and affects firms’ target prices. It does not mechanically de-
 70 termine whether the dividend appears as lower prices, higher wages, retained

71 profits, shorter work hours, or unemployment. Consistent with the Schum-
72 peterian emphasis on innovation-driven firm turnover (Schumpeter, 1942),
73 the model includes firm entry and exit as well as a solo-enterprise channel.
74 However, it does not yet include an empirically calibrated task–skill exposure
75 matrix or endogenous skill accumulation. It is therefore a transition stress
76 test with firm turnover, not a complete long-run equilibrium model.

77 At the macroeconomic level, AI can expand productive capacity while
78 weakening labor income and effective demand, so employment effects depend
79 on demand creation, task reinstatement, and the distribution of productivity
80 rents (Bessen, 2019; Berg et al., 2018; Acemoglu, 2025). Empirical evidence
81 on automation also shows that local labor-market adjustment can remain un-
82 even even when aggregate productivity rises (Acemoglu and Restrepo, 2020).
83 These mechanisms motivate a model in which technical capability does not
84 predetermine the division of gains among prices, wages, profits, shorter hours,
85 and employment.

86 1.2.3. Fiscal stabilization, distribution, and agent-based evidence

87 Taxes, transfers, procurement, and public services can absorb or redi-
88 rect demand shocks (Auerbach and Feenberg, 2000; McKay and Reis, 2016).
89 Distribution is evaluated with bottom-60% real consumption and Atkinson
90 indices (Atkinson, 1970). The agent-based design follows the tradition of
91 stock-flow-aware macroeconomic simulation (Dosi et al., 2010; Caiani et al.,
92 2016; Dawid and Delli Gatti, 2018). LLMs add bounded adaptive choices,
93 but their natural-language outputs are not treated as economic facts. Ac-
94 counting, prices, employment totals, taxes, credit, and balance sheets remain
95 under deterministic code.

96 The model also connects heterogeneous-agent macroeconomics (Heath-
97 cote, 2005; Kaplan et al., 2018; Dosi et al., 2013, 2022) with emerging LLM-
98 based social simulation. Generative agents have been used to represent in-
99 dividual and collective behavior in controlled social environments (Argyle
100 et al., 2023; Park et al., 2023; Bail, 2024; Gao et al., 2024), while AgentSoci-
101 ety provides infrastructure for large-scale and executable social experiments
102 (Piao et al., 2025, 2026). Our design differs by restricting LLMs to pre-
103 specified categorical stances and placing all monetary, labor-market, credit,
104 and fiscal settlement in deterministic code. The intended contribution is
105 therefore not to treat LLM outputs as calibrated human behavior, but to
106 test whether bounded adaptive choices alter the transition paths generated
107 by a transparent economic mechanism.

108 2. Empirical Anchors and Parameterization

109 Table 1 maps each external indicator to its baseline-model use. It distin-
 110 guishes direct statistical anchors, ratios derived from official totals, literature
 111 or regulatory benchmarks, and experimental settings.

Table 1: Empirical anchors, benchmark indicators, and baseline-model mappings.

Domain	External indicator and source	Baseline-model mapping	Evidential use
Income level and distribution	2025 per-capita disposable income is CNY 43,377; income-quintile means are CNY 10,150, 22,702, 35,536, 55,586, and 103,778 (National Bureau of Statistics of China, 2026a,b).	Annual disposable income for the five equally sized household groups.	Direct statistical anchor.
Income composition	2025 per-capita wage income is CNY 24,555, or 56.6% of disposable income (National Bureau of Statistics of China, 2026a).	Income-weighted group net-wage shares reproduce 56.64%.	Direct structural anchor.
Household consumption	2025 per-capita consumption is CNY 29,476; the expenditure-to-income ratio is 0.680 (National Bureau of Statistics of China, 2026a).	Annual consumption level and group consumption propensities of 0.55–0.98.	Direct total; group propensities are calibrated.
Macro household accounts	2024 income, consumption, population, and GDP imply household-income/GDP of 0.431 and household-saving/GDP of 0.137 (National Bureau of Statistics of China, 2025a,b).	Reference ratios are 0.43 and 0.137.	Derived official-data benchmark.

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Domain	External indicator and source	Baseline-model mapping	Evidential use
Labor market	The 2025 average urban surveyed unemployment rate is 5.2% (National Bureau of Statistics of China, 2026b).	Initial and target unemployment rates are both 0.052.	Direct statistical anchor.
Credit cost	In March 2025, the new-loan and corporate-loan weighted rates were 3.44% and 3.26% (People’s Bank of China, 2025).	The annual firm-loan rate is 4.0%.	Market benchmark with a conservative model premium.
Bank resilience	At end-2025, commercial-bank capital adequacy was 15.46% and the non-performing-loan ratio was 1.50%; the minimum capital requirement is 8% plus a 2.5% conservation buffer (National Financial Regulatory Administration, 2023, 2026).	Minimum capital adequacy is 12%; baseline annual default probability is 1.5%.	Capital adequacy is a prudential benchmark. The 1.5% default probability is an independent experimental setting, not a mapping from the NPL ratio.
Fiscal space	The 2025 national budget set the deficit ratio at about 4% (State Council of the People’s Republic of China, 2025).	The baseline deficit limit is 4%; the constrained case uses 2%.	National fiscal benchmark, not a city estimate.
AI productivity	Workplace evidence reports a 15% average productivity gain in a specific customer-support setting (Brynjolfsson et al., 2025).	Firm targets are 5%, 14%, and 20%; personal AI adds 8%.	Experimental heterogeneity centered on a literature benchmark, not an economy-wide estimate.

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Table 1 continued from the previous page.

Domain	External indicator and source	Baseline-model mapping	Evidential use
Other behavioral and policy settings	No direct official counterpart is imposed.	Firm scale and composition, liquidity buffers, tax and price pass-through, debt limits, adjustment speeds, levy recycling, and entry/exit rules.	Experimental settings tested through scenarios and sensitivity analysis.

Notes: Direct anchors set selected baseline levels. Derived benchmarks are calculated from official aggregates. Literature and regulatory benchmarks set plausible ranges. Experimental settings are not empirical estimates. National indicators provide scale and consistency checks; they do not make the model a calibrated representation of a particular city.

112 All monetary flows are represented in monthly CNY model units. Annual
 113 per-capita household anchors are divided by 12 and assigned to 500 house-
 114 hold decision units in five equally sized income groups. Aggregate monetary
 115 results therefore scale with the simulated population and should not be in-
 116 terpreted as estimates for an actual city. Parameters not directly anchored
 117 by Table 1 are pre-specified experimental settings whose values and sources
 118 are summarized in the model description and [Appendix A](#).

119 3. Methodology

120 This section develops the model through seven linked components. Sec-
 121 tion [3.1](#) defines the closed-loop monthly urban economy, and Section [3.2](#)
 122 specifies its sequence of decisions, market allocation, and accounting settle-
 123 ment. Section [3.3](#) formalizes household income, consumption, and liquidity;
 124 Section [3.4](#) describes firm production, pricing, and employment; Section [3.5](#)
 125 sets out government policy responses and fiscal constraints; and Section [3.6](#)
 126 defines bank intermediation and credit limits. Finally, Section [3.7](#) explains
 127 how bounded LLM-generated categorical decisions are translated into deter-
 128 ministic economic actions and auditable state updates.

129 3.1. Closed-loop stylized urban economy

Table 2: Core behavioral and accounting parameters used in the model equations.

Symbol	Meaning	Baseline value	Status
δ_U, κ_C	Unemployment and duration-dependent consumption penalties	0.12, 0.10	Pre-specified
λ	Maximum liquid-asset drawdown share	0.35	Pre-specified
c^{AI}	Personal-AI monthly cost	CNY 30	Pre-specified
$\bar{u}^{\text{cap}}, \kappa_Q$	Target capacity utilization and productivity scale	0.85, 3,343	Pre-specified
ν_A, ν_D, ν_p	AI-adoption, demand-expectation, and price-adjustment speeds	0.10, 0.30, 0.20	Pre-specified
ϕ_w, ϕ_p	Wage and price pass-through shares of the AI productivity gain	0.20, 0.55	Pre-specified
$\bar{u}^U, \ell_G$	Target unemployment rate and government policy lag	0.052, 2 months	Statistical anchor; pre-specified lag
ρ_T, ρ_G	Transfer and procurement response rates	0.50, 0.30	Pre-specified
ρ_D, ρ_M	Deposit-reserve and managed-fund liquidity ratios	0.10, 0.05	Prudential settings
$\bar{s}_F, \bar{\ell}$	Firm-loan portfolio and total loan-to-funding limits	0.58, 0.80	Prudential settings
$\nu_B, \underline{\kappa}, \omega_F$	Credit-adjustment speed, minimum capital adequacy, and firm-loan risk weight	0.25, 0.12, 1.00	Mixed

Notes: “Pre-specified” means fixed in the baseline configuration before the retained scenario runs; it does not imply external preregistration. Scenario-specific institutional parameters are reported in [Appendix A](#).

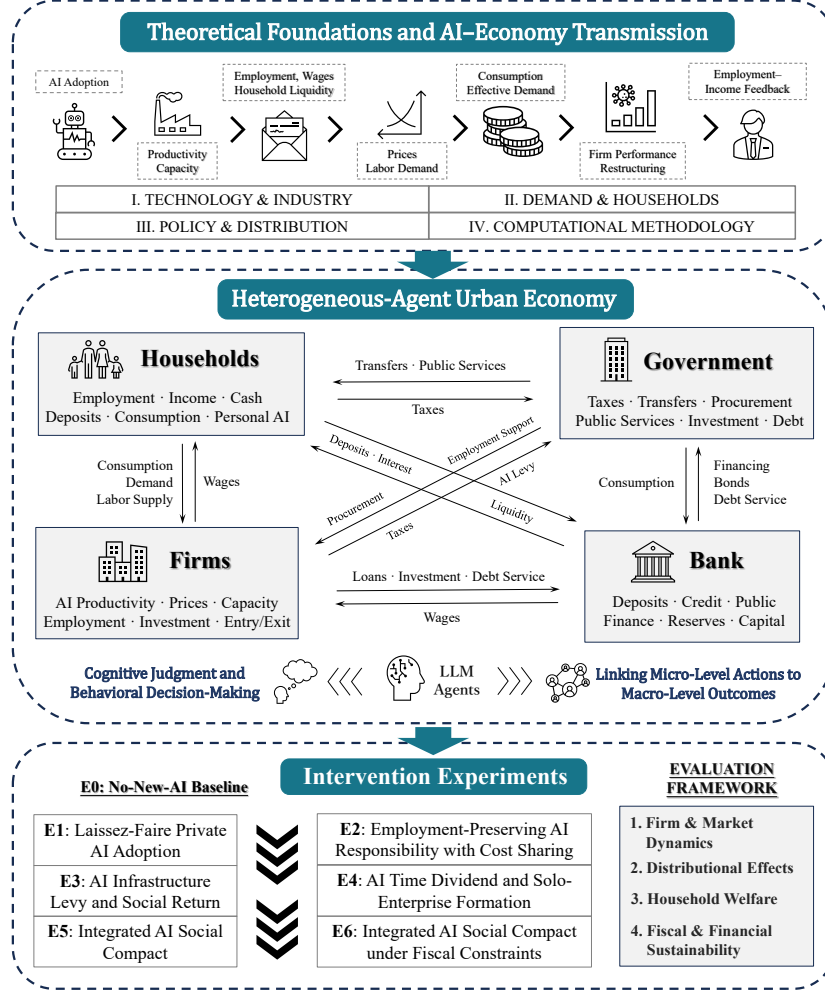


Figure 1: Urban Cup conceptual framework. The theoretical transition layer motivates a heterogeneous-agent stylized urban economy in which households, firms, government, and the banking sector interact. The E0–E6 experiments modify institutional channels, while evaluation spans firm and market dynamics, distribution, household welfare, fiscal and financial sustainability, and robustness.

130 The model is a monthly stock–flow-aware simulation of a stylized urban
 131 economy. Its state at the end of month t is

$$\mathcal{S}_t = (\{H_{i,t}\}_{i=1}^N, \{F_{f,t}\}_{f=1}^{J_t}, \mathcal{G}_t, B_t), \quad (1)$$

132 where $H_{i,t}$, $F_{f,t}$, $\mathcal{G}_t$, and B_t denote household, firm, government, and bank
 133 states, respectively. The baseline contains $N = 500$ household decision units,
 134 approximately 30 incumbent firms, one government, and one banking sector.
 135 A household is both a consumption unit and a potential labor supplier; it
 136 is not intended to represent every member of a literal family. Households
 137 belong to five income groups and differ in productivity, wages, consumption
 138 propensities, and liquid-asset buffers.

139 Firms differ along three dimensions: size, exposure to AI, and organiza-
 140 tional culture. AI-high, AI-augmented, and human-service firms have differ-
 141 ent target productivity multipliers. Augmentation, cost-cutting, and adap-
 142 tive cultures differ in employee-retention commitments, adjustment speeds,
 143 complementary-job creation, and price pass-through. The government col-
 144 lects labor, indirect, profit, solo-enterprise, and AI-related charges and allo-
 145 cates resources to transfers, procurement, public services, public investment,
 146 government AI, and temporary retention subsidies. The bank accepts house-
 147 hold deposits and managed funds, supplies investment credit to firms, pur-
 148 chases government debt, and maintains reserves, provisions, and equity. The
 149 bank is an economic entity but not an LLM decision maker: its transactions
 150 are governed entirely by deterministic balance-sheet and prudential rules.

151 3.2. *Within-month sequence and market settlement*

152 The order of operations is fixed so that behavioral decisions cannot over-
 153 write economic accounting. At the start of a month, the engine applies the
 154 bounded household, firm, and government intentions submitted for that pe-
 155 riod. It then updates AI adoption, firm prices, solo-enterprise status, bank
 156 funding and investment credit, and productivity-linked wages. The govern-
 157 ment forms a fiscal plan from lagged macroeconomic observations. House-
 158 holds subsequently receive wages, other income, bank returns, and trans-
 159 fers and submit consumption orders. Household demand, government pur-
 160 chases, firm investment, and external solo-enterprise demand are then allo-
 161 cated across suppliers. After goods-market settlement, the engine updates
 162 household cash, firm profits, taxes, levies, government accounts, and bank
 163 loans. Firms finally revise expected demand, adjust employment, and enter
 164 or exit before capacities and market shares are carried into the next month.

Let $C_{i,t}^p$ denote planned nominal household spending, G_t^S public services, G_t^P procurement, G_t^{AI} government AI spending, I_t^G public investment, $I_{f,t}^F$ firm investment, and X_t^{solo} net-new demand for solo-enterprise output. The latter contains household cash-financed induced demand X_t^{ind} and demand from outside the modeled economy X_t^{ext} , so that $X_t^{solo} = X_t^{ind} + X_t^{ext}$. Substitution from household and government orders and business-to-business demand from firm investment reallocate existing orders and are not added again. Total orders are

$$O_t = \sum_i C_{i,t}^p + G_t^S + G_t^P + G_t^{AI} + I_t^G + \sum_f I_{f,t}^F + X_t^{solo}. \quad (2)$$

Orders to an incumbent firm combine its market-share allocation with any targeted employment-support procurement. If $O_{f,t}$ is the resulting order, nominal sales are constrained by productive capacity:

$$S_{f,t} = \min \{O_{f,t}, p_{f,t} \bar{Q}_{f,t}\}. \quad (3)$$

The common order-fulfilment rate is

$$\omega_t = \min \left\{ 1, \frac{\sum_f S_{f,t} + S_t^{solo}}{O_t} \right\}, \quad (4)$$

and scales realized household spending, government purchases, and investment. This rationing rule closes the goods market without allowing agents to create sales or fiscal resources directly.

3.3. Household income, consumption, and liquidity

Before current solo-enterprise income is settled, disposable resources for household i are

$$Y_{i,t}^d = (1 - \tau_L) e_{i,t} w_{i,t} + y_{i,t}^o + r_{i,t}^D + TR_{i,t}, \quad (5)$$

where $e_{i,t}$ is employment status, τ_L is the labor-income tax rate, $w_{i,t}$ is gross wage income, $y_{i,t}^o$ is other baseline income, $r_{i,t}^D$ is income from bank deposits and managed funds, and $TR_{i,t}$ is the unemployment transfer. Other income is partially reduced after a model-generated displacement, while the transfer replaces part of the last net wage subject to a minimum benefit.

Households follow a buffer-stock consumption rule rather than solving a forward-looking intertemporal optimization problem. Let $C_{i,0}$ and $Y_{i,0}^d$ be

the household's calibrated consumption and disposable-income anchors, m_i its marginal propensity to consume, $d_{i,t}^U$ the duration of model-generated unemployment, and $\mathbb{I}_{i,t}^U$ an indicator for that state. Desired non-AI spending is

$$\tilde{C}_{i,t} = s_{i,t}^C \left[\{C_{i,0} + m_i (Y_{i,t}^d - Y_{i,0}^d)\} (1 - \delta_U \mathbb{I}_{i,t}^U) - \kappa_C C_{i,0} \min \left(\frac{d_{i,t}^U}{6}, 1 \right) \right], \quad (6)$$

where $s_{i,t}^C \in \{1.00, 0.94, 0.86\}$ represents normal, cautious, and defensive consumption stances. The unemployment penalty δ_U and the duration-dependent precautionary term κ_C distinguish a temporary income loss from persistent displacement.

Minimum consumption places a floor on desired non-AI spending. Actual planned spending may nevertheless fall below that floor when current income and the permitted draw on liquid assets are insufficient:

$$C_{i,t}^p = \min \left\{ \max \left(C_i^{\min}, \tilde{C}_{i,t} \right) + c^{AI} a_{i,t}, \right. \\ \left. Y_{i,t}^d + \lambda (M_{i,t-1} + D_{i,t-1}) \right\}, \quad \lambda = 0.35, \quad (7)$$

where $M_{i,t-1}$ and $D_{i,t-1}$ are cash and deposits, $a_{i,t}$ is personal-AI use, and c^{AI} is its monthly cost. Personal-AI spending is available only to employed households when private AI is active. Deposits are withdrawn when current cash is insufficient. After market rationing, personal-AI expenditure is excluded from ordinary real consumption:

$$C_{i,t}^{real} = \frac{\omega_t (C_{i,t}^p - c^{AI} a_{i,t})}{P_t}, \quad (8)$$

where P_t is a fixed-base-quantity Laspeyres price index. Residual cash above an income- and unemployment-adjusted target is partly swept into bank deposits and managed funds. In the solo-enterprise scenarios, eligible households may leave wage employment, earn business income, and re-enter unemployment after persistent losses; wage employment and self-employment are mutually exclusive.

3.4. Firm production, pricing, and employment

After the common pre-equilibrium, the AI multiplier of firm f converges gradually to its type-specific target A_f^* :

$$A_{f,t} = A_{f,t-1} + \nu_A (A_f^* - A_{f,t-1}). \quad (9)$$

215 For employees of firm f , part of the productivity gain passes into wages:

$$w_{i,f,t} = w_{i,0} [1 + \phi_w (A_{f,t} - 1)] . \quad (10)$$

216 The no-levy target price passes another portion of the gain to buyers. With
217 an AI levy, a configurable share of the unit charge is added to that target:

$$\begin{aligned} p_{f,t}^{*,0} &= \max \{p_{\min}, 1 - \phi_p \chi_f (A_{f,t} - 1)\} , \\ p_{f,t}^* &= p_{f,t}^{*,0} + \zeta_p \ell_{f,t}^{AI}, \quad p_{f,t} = p_{f,t-1} + \nu_p (p_{f,t}^* - p_{f,t-1}) , \end{aligned} \quad (11)$$

218 where χ_f is the culture-specific price aggressiveness, $\ell_{f,t}^{AI}$ is the unit levy, and
219 ζ_p is the consumer pass-through rate.

220 Effective worker productivity is $\tilde{z}_{i,t} = z_i(1 + g_i a_{i,t})$, where g_i is the pro-
221 ductivity gain from personal AI. Firm capacity is

$$\bar{Q}_{f,t} = \kappa_Q A_{f,t} \sum_{i \in \mathcal{L}_{f,t}} \tilde{z}_{i,t} \left[1 + \alpha_K \frac{K_{f,t}}{p_{f,0} \bar{Q}_{f,0}} \right] , \quad (12)$$

222 where $\mathcal{L}_{f,t}$ is the workforce and $K_{f,t}$ is loan-financed productive capital after
223 depreciation.

224 Expected nominal demand follows an adaptive rule:

$$D_{f,t}^e = D_{f,t-1}^e + \nu_D (S_{f,t-1} - D_{f,t-1}^e) . \quad (13)$$

225 The continuous labor requirements with and without firm AI are

$$L_{f,t}^{AI} = \frac{D_{f,t}^e}{\bar{u}^{\text{cap}} \kappa_Q A_{f,t} \tilde{z}_{f,t} p_{f,t}}, \quad L_{f,t}^0 = \frac{D_{f,t}^e}{\bar{u}^{\text{cap}} \kappa_Q \bar{z}_{f,t}^0 p_{f,t}}, \quad (14)$$

226 where $\bar{u}^{\text{cap}} = 0.85$ is target capacity utilization. The discrete employment
227 target combines rounded demand, AI-complementary jobs, and the culture-
228 specific retention floor. Under employment responsibility, it is also bounded
229 below by a configurable share of $L_{f,t}^0$, unless the firm has remained distressed
230 long enough to qualify for an exemption. Adjustment toward the target is
231 partial; patient and aggressive stances multiply the baseline speed by 0.60
232 and 1.40, respectively. Protected workers remain employed but their required
233 hours fall with $L_{f,t}^{AI}/L_{f,t}$. This mechanism therefore converts some AI labor
234 savings into shorter work time rather than mechanically fixing output or
235 employment.

236 Market shares respond to productive capacity, relative prices, and em-
 237 ployment retention. Before normalization, firm attractiveness is

$$\mathcal{A}_{f,t} = \left(\frac{\bar{Q}_{f,t}}{\sum_j \bar{Q}_{j,t}} \right)^{\eta_Q} \left(\frac{\bar{p}_t}{p_{f,t}} \right)^{\eta_p} \left[\max \left(\frac{L_{f,t}}{L_{f,0}}, 0.05 \right) \right]^{\eta_L}. \quad (15)$$

238 Below-cost pricing may reduce this attractiveness. Normalization gives the
 239 desired share

$$s_{f,t}^* = \frac{\mathcal{A}_{f,t}}{\sum_j \mathcal{A}_{j,t}},$$

240 and actual shares move only partly toward $s_{f,t}^*$ each month.

241 After sales, operating cash profit before loan service and depreciation is

$$\Pi_{f,t} = S_{f,t} - W_{f,t} - F_f - \tau_S S_{f,t} - L_{f,t}^{levy} + Sub_{f,t}^{ret}, \quad (16)$$

242 The model's retained operating profit after profit tax is

$$\Pi_{f,t}^R = \Pi_{f,t} - \tau_\pi \max(\Pi_{f,t}, 0). \quad (17)$$

243 Loan interest and principal are settled separately against firm cash after
 244 Eq. (17). Accordingly, $\Pi_{f,t}^R$ is a model cash measure rather than accounting
 245 net income; in particular, loan interest is not deducted from the model's
 246 profit-tax base. Persistent negative cash, repeated operating losses, or deep
 247 workforce contraction can trigger exit. Employment responsibility provides
 248 a short restructuring grace period but does not cancel bankruptcy. Entry
 249 requires unemployed workers, an AI or unmet-demand opportunity, and a
 250 bank-financed start-up loan; entrants recruit the most productive available
 251 workers.

252 3.5. Government behavior and fiscal constraints

253 The government observes consumption and unemployment with a two-
 254 month policy lag. Its demand and labor-market gaps are

$$g_t^C = \max(C^0 - C_{t-\ell_G}, 0), \quad g_t^U = \max(u_{t-\ell_G} - \bar{u}^U, 0), \quad \ell_G = 2. \quad (18)$$

255 Here C^0 is baseline aggregate consumption, u_t is the unemployment rate,
 256 and $\bar{u}^U = 0.052$ is the policy target unemployment rate. When the relevant
 257 channels are active, the desired extra transfer pool and regular procurement
 258 response are

$$TR_t^{extra} = \rho_T g_t^U N \bar{w}_t, \quad G_t^{P,*} = G^{P,0} + \rho_G g_t^C + G_t^{emp} + G_t^{div}, \quad (19)$$

259 where G_t^{emp} targets firms that retain at least 85% of their initial workforce
 260 and G_t^{div} may recycle purchasing-power gains. Policy changes are gradual
 261 rather than instantaneous. The government stance further scales procure-
 262 ment: stabilization and balanced-support raise it modestly, whereas fiscal-
 263 guard reduces discretionary support.

264 Total tax and levy revenue is

$$T_t = \tau_L W_t + \tau_S S_t + \tau_\pi \sum_f \max(\Pi_{f,t}, 0) + T_t^{solo} + L_t^{levy}. \quad (20)$$

265 Before expenditure is authorized, the engine computes borrowing room:

$$BR_t = \min \left\{ \delta Y^0, [\bar{d}(12Y^0) - (D_{t-1}^G + A_{t-1}^G)]_+, BR_t^{bank} \right\}, \quad (21)$$

266 where δ is the annual-deficit parameter applied as a monthly flow cap, $\bar{d}$ is
 267 the debt-limit ratio, D^G is formal government debt, and A^G denotes arrears.
 268 Available fiscal resources equal current cash, 95% of the previous observed
 269 tax flow, and BR_t . If planned expenditure exceeds these resources, scenario-
 270 specific cut priorities reduce discretionary components. Base unemployment
 271 transfers and a model-imposed public-service floor are protected; any residual
 272 gap is reported as arrears rather than silently removing the obligation.

273 Let E_t^G denote total realized government expenditure. The fiscal balance
 274 is $FB_t = T_t - E_t^G$. A surplus first repays arrears and formal debt and then
 275 accumulates as cash. A deficit uses cash, eligible bank borrowing, and fi-
 276 nally arrears. Public-service output depends on real service spending, lagged
 277 government-AI use, and accumulated public capital:

$$PSI_t = \frac{G_t^S P_t^{-1} (1 + \psi_A a_{t-\ell_S}^G)}{G_0^S (1 + \psi_A a_0^G)} \left[1 + \alpha_G \frac{K_t^G}{12G_0^S} \right]. \quad (22)$$

278 The AI-levy account is separately auditable: realized levy revenue increases
 279 its balance, while earmarked public-service and public-investment spending
 280 reduce it.

281 3.6. Bank intermediation and financial constraints

282 The deterministic bank links household saving, firm investment, and gov-
 283 ernment finance. Its accounting identity is

$$R_t + L_t^F + L_t^G + OFA_t - Prov_t = Dep_t + MF_t + E_t, \quad (23)$$

284 where R is reserves, L^F firm loans, L^G government loans or bonds, OFA
 285 other financial assets, $Prov$ loan-loss provisions, Dep household deposits,
 286 MF managed funds, and E bank equity.

287 The aggregate amount available for new firm credit is the minimum of
 288 four positive-part constraints:

$$\mathcal{C}_t^F = \min \left\{ \begin{array}{l} \nu_B [R_t - \rho_D Dep_t - \rho_M MF_t]_+, \\ [\bar{s}_F (Dep_t + MF_t) - L_t^F]_+, \\ [\bar{\ell} (Dep_t + MF_t) - L_t^F - L_t^G]_+, \\ \left[\frac{E_t / \underline{\kappa} - RWA_t}{\omega_F} \right]_+ \end{array} \right\}. \quad (24)$$

289 These terms impose liquidity reserves, a firm-loan portfolio limit, a total
 290 loan-to-funding limit, and a minimum capital-adequacy ratio $\underline{\kappa}$, respectively.
 291 Firm loan requests rise with replacement investment and capacity utilization
 292 above 80%. Approval additionally requires the request to satisfy borrower
 293 concentration, leverage relative to baseline sales, debt-service coverage, and
 294 default-probability thresholds. If eligible requests exceed $\mathcal{C}_t^F$, credit is allo-
 295 cated proportionally.

296 Firm repayments restore bank reserves and reduce outstanding loans; in-
 297 terest raises bank equity. Delinquency moves loans through normal, watch,
 298 substandard, doubtful, and loss categories, each with a corresponding provi-
 299 sion rate. Government borrowing competes for the same funding and capital
 300 space and is limited by the bank’s maximum government-bond share. The
 301 bank therefore transmits household liquidity and fiscal borrowing into invest-
 302 ment and entry rather than serving as a passive residual account.

303 3.7. Bounded LLM participation and deterministic settlement

304 The hybrid architecture separates interpretation from settlement. When
 305 enabled, households may choose only normal, cautious, or defensive consump-
 306 tion stances; firms may choose patient, baseline, or aggressive labor stances;
 307 and government may choose baseline, stabilization, balanced-support, or
 308 fiscal-guard stances. Structured outputs are parsed against these whitelists.
 309 Invalid or missing outputs revert to the pre-specified rule-based action and
 310 are recorded in the decision audit.

311 The retained runs used the HKUST OpenAI-compatible endpoint and
 312 requested the provider alias `gpt-3.5-turbo`; response metadata identifies

gpt-4o-mini-2024-07-18 under the documented provider-alias policy. Requests were batched in groups of 16 with at most four concurrent calls. Prompt templates, categorical schemas, parsed responses, fallback flags, and response-model identifiers are retained in the run artifacts. Generation on the provider side is not seed-deterministic, so exact behavioral replay requires frozen response caches in addition to the economic seed.

No LLM can directly choose consumption expenditure, wages, prices, employment levels, tax rates, loans, government debt, or balance-sheet entries. Its categorical stance modifies a small set of adjustment multipliers, after which the deterministic engine applies Eqs. (2)–(24) and records the complete economic state.

The retained matrix uses R1, in which only government uses an LLM, and R2, in which firms and government use LLMs while households remain rule-driven. The bank is deterministic in both decision architectures. R2 is the main experiment because firm labor adjustment and survival are central to the transition mechanism; R1 is an ablation that identifies whether those paths depend on enterprise-level LLM decisions rather than on a change in accounting or policy parameters.

4. Simulation Design

4.1. Initialization and common pre-equilibrium

Initial unemployment is 5.2%. Within-group productivity is sampled from a normalized lognormal distribution. Each retained path contains 24 common pre-equilibrium months. The state must pass accounting, boundary, and warm-up stability checks before all scenarios fork from the same month-24 state. Private AI and institutional treatments activate in month 25.

The warm-up gate requires all 24 observations, accounting errors within pre-specified absolute and relative tolerances, bounded numeric states, flow drift over the final 12 warm-up months no greater than 0.5%, and employment drift no greater than 0.5 percentage points (or one household unit). The common checkpoint, source fingerprint, economic seed, and pre-treatment histories are identical across E0–E6 within each decision architecture. This common-state design follows the logic of common random numbers for simulation comparisons (Kleijnen, 1988); the model description is organized to make agents, state variables, scheduling, and process rules auditable in the spirit of the ODD protocol (Grimm et al., 2006).

348 *4.2. Institutional scenarios*

349 Table 3 summarizes the primary analytical purpose of each institutional
350 branch.

Table 3: Primary analytical purposes of the institutional scenarios.

Scenario	Primary exploratory purpose
E0	Establish the post-warm-up time counterfactual without new private AI adoption.
E1	Isolate the transition effects of laissez-faire private AI relative to E0.
E2	Test whether employment responsibility and temporary wage-cost sharing convert AI labor savings into lower work intensity rather than layoffs.
E3	Test whether an AI-rent levy and earmarked recycling accelerate the public return from private AI gains.
E4	Test whether AI-saved time supports solo-enterprise formation and shifts labor from wage employment toward self-employment.
E5	Test the joint operation of employment responsibility, levy-funded public return, solo enterprise, and active demand as an integrated AI social compact.
E6	Assess how tighter deficit and debt limits alter the operation of the integrated compact.

Notes: All scenarios fork from the same qualified month-24 state and activate in month 25. Appendix Table A.4 reports the pre-specified mechanisms, parameter values, fiscal limits, and counterfactual contrasts.

351 E0 is not a failed policy; it is the post-warm-up time counterfactual with-
352 out new private AI. E1 identifies the private-AI shock. E2–E4 each add one
353 mechanism and must be interpreted against E1. E5 evaluates the portfolio
354 rather than a sum of independent effects. The E6–E5 contrast changes the
355 fiscal constraints, although exact numerical attribution also requires common
356 or frozen LLM responses.

357 4.3. Outcome measures

358 Demand outcomes include cumulative real consumption, bottom-60% real
359 consumption, and disposable income over months 25–120. Labor outcomes
360 include peak unemployment, tail employment, self-employment, blocked AI-
361 attributable layoffs, exit-job losses, entry jobs, and required work hours. So-
362 cial outcomes include liquidity vulnerability, essential-cash shortfall, public
363 services, and Atkinson inequality with inequality-aversion parameter $\epsilon = 1$.
364 Firm and fiscal outcomes include entry, exit, market concentration, levy rev-
365 enue and use, curtailment, arrears, and debt. “Cumulative” denotes a sum
366 over months 25–120, “peak” the post-shock maximum, “tail” the mean over
367 months 109–120, and “ending” the month-120 value. Maximum debt ratios
368 are calculated over months 25–120.

369 5. Main Findings

370 This section reports six linked findings. Section 5.1 contrasts productivity
371 gains with transition stress; Section 5.2 identifies exit leakage; Section 5.3
372 traces levy-funded public services; Section 5.4 separates solo enterprise from
373 net-new demand; Section 5.5 assesses joint policy under fiscal constraints;
374 and Section 5.6 tests sensitivity to LLM-participation architecture.

375 5.1. Private AI shock and policy trade-offs

376 This subsection first compares E0 with E1 under R2 to isolate the aggre-
377 gate gains and transition costs of laissez-faire private AI. It then compares
378 E2–E6 with E1 to assess how alternative institutions redistribute those gains
379 and costs. Figure 2 reports endpoint outcomes for consumption, unemploy-
380 ment, liquidity vulnerability, public services, firm survival, and public debt,
381 while Figure 3 traces monthly consumption and disposable-income inequality.

382 Moving from E0 to E1 raises cumulative real consumption from approx-
383 imately CNY 123.4 million to CNY 130.1 million (+5.4%) and bottom-60%
384 cumulative real consumption by 5.6%. However, peak unemployment rises
385 from 2.8% to 13.0%, and tail liquidity vulnerability rises from 1.20% to 8.92%.
386 Aggregate productivity and demand gains therefore coexist with substantial
387 employment and household-liquidity stress.

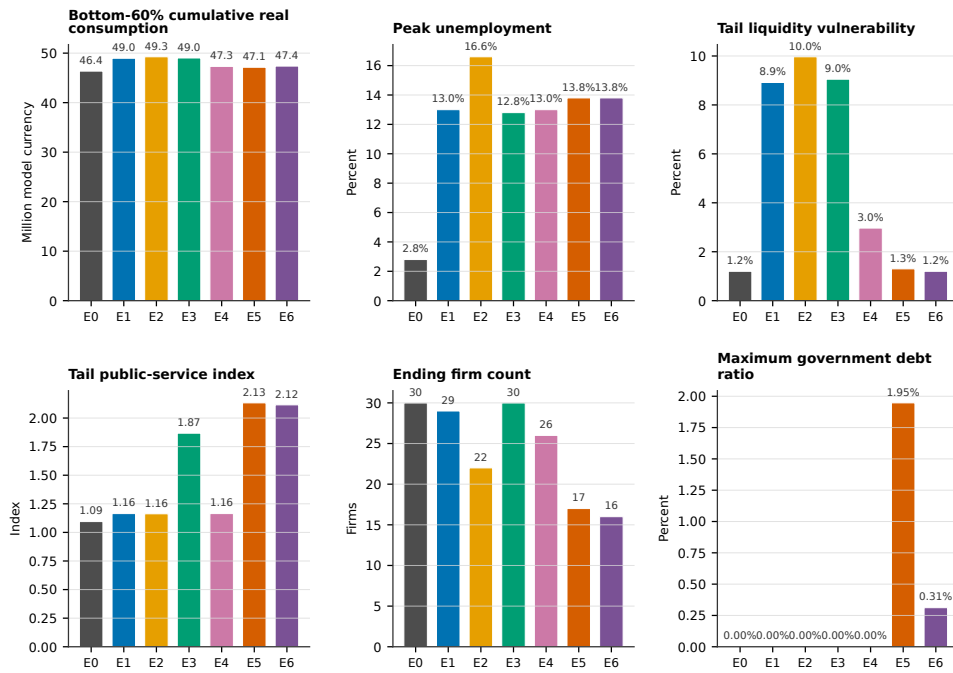


Figure 2: Core policy trade-offs in R2 across E0–E6. Panels compare bottom-60% cumulative real consumption, peak unemployment, tail liquidity vulnerability, the tail public-service index, ending firm count, and the maximum government-debt ratio.

388 The policy variants redistribute rather than eliminate transition costs.
389 E3 remains close to E1 in cumulative consumption and peak unemployment
390 while raising public services. E4–E6 reduce liquidity vulnerability but lower
391 consumption, firm survival, or both. No scenario therefore dominates across
392 all outcomes. These differences also develop over time: Figure 3 traces the
393 divergence in monthly consumption and disposable-income inequality.

394 Taken together, the results show why aggregate consumption alone is an
395 insufficient criterion for AI governance. Institutional alternatives must also
396 be evaluated by how they distribute productivity gains and transition losses
397 across employment, household liquidity, income distribution, firm survival,
398 public services, and fiscal exposure.

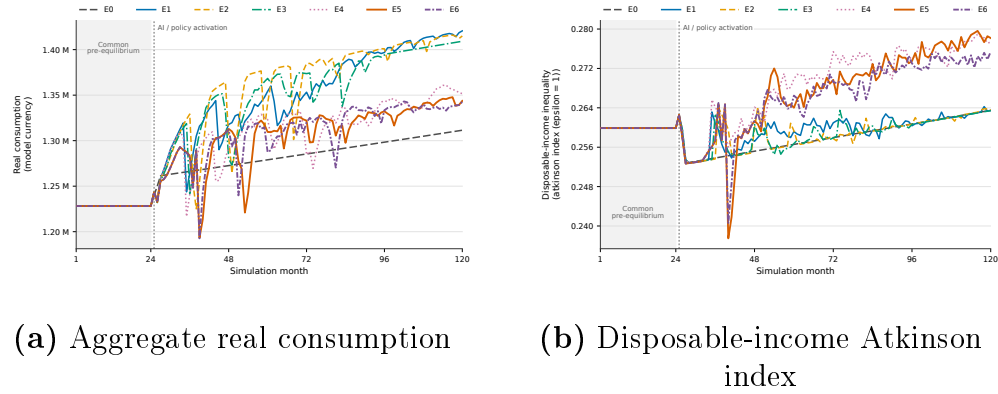


Figure 3: Dynamic demand and distribution effects in R2 across E0–E6. Monthly real consumption traces aggregate demand, while the disposable-income Atkinson index tracks distributional divergence.

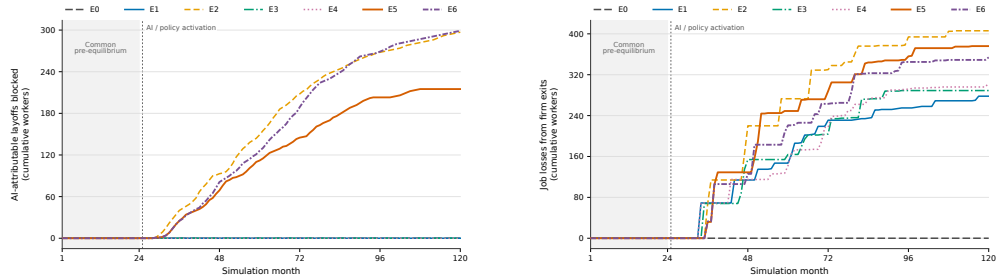
399 5.2. *Employment responsibility: retention and exit leakage*

400 This subsection compares E2 with E1 under R2 to test whether employ-
401 ment responsibility preserves jobs without shifting labor losses to firm exit.
402 Figure 4 contrasts cumulative AI-attributable layoffs blocked inside surviving
403 firms with job losses caused by firm closures.

404 E2 blocks 297 AI-attributable layoffs, reduces tail required work hours
405 from 160 to 147.8, and provides approximately CNY 51,200 in temporary
406 wage-cost sharing. Cumulative consumption is 0.60% above E1. The rule
407 therefore activates as intended and retains workers inside operating firms.

408 This consumption improvement coexists with a deterioration in firm sur-
 409 vival. Firm exits remove 406 jobs, new firms create only 56, the ending firm
 410 count falls to 22, and peak unemployment reaches 16.6%, 3.6 percentage
 411 points above E1. The comparison reveals the policy boundary: protection
 412 against layoffs within surviving firms does not cover liquidation or weak job
 413 creation.

414 Employment responsibility must therefore be paired with measures that
 415 support firm survival, restructuring, and new-job creation; otherwise, em-
 416 ployment losses can move from direct layoffs to enterprise exit.



(a) AI-attributable layoffs blocked

(b) Job losses from firm exits

Figure 4: Employment retention and exit leakage in R2. Panel (a) tracks AI-attributable layoffs blocked inside surviving firms, while panel (b) tracks job losses from firm exits that fall outside the retention rule.

417 5.3. AI levy and social return

418 This subsection compares E3 with E1 under R2 to test whether an AI
 419 levy can convert part of the private productivity gain into public services
 420 without materially weakening consumption, employment, or firm survival.
 421 Figure 5 links cumulative levy receipts to the resulting public-service path.

422 E3 collects CNY 5.157 million in AI infrastructure levy revenue. Approx-
 423 imately 70% finances public services and 30% finances public investment.
 424 Relative to E1, cumulative consumption is 0.09% higher, peak unemploy-
 425 ment is 0.2 percentage points lower, the tail public-service index is 60.4%
 426 higher, and 30 firms remain.

427 Among the current scenarios, E3 therefore shows the most favorable qual-
 428 itative combination of aggregate demand, labor-market stress, public ser-
 429 vices, and firm survival across the reported outcomes. This is not a weighted

multi-objective ranking. The result demonstrates that the programmed levy-to-service channel operates as specified, but it does not identify an optimal levy rate or an externally valid public-service multiplier.

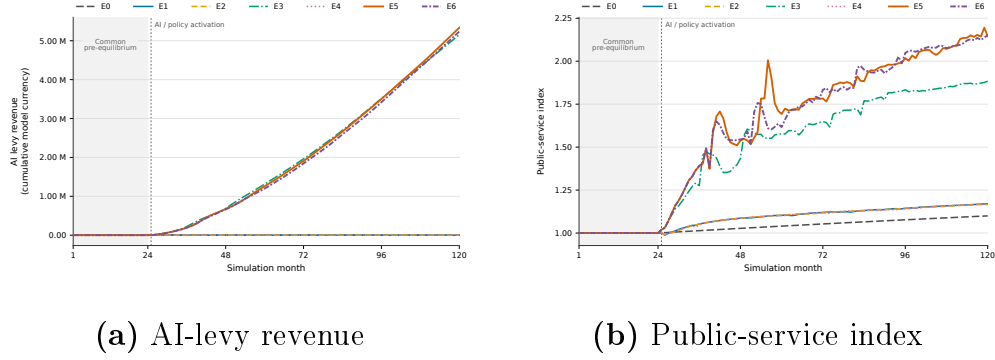


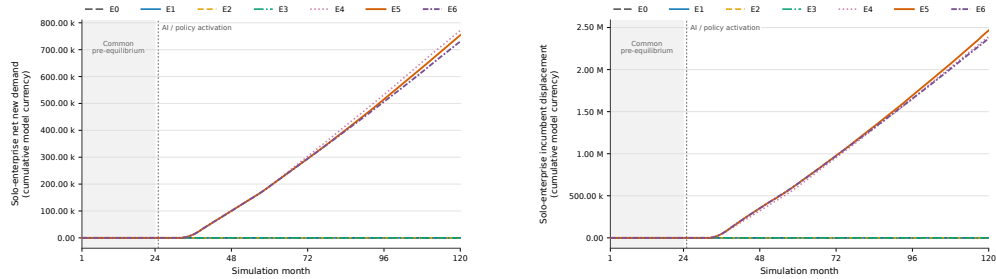
Figure 5: Levy-to-service transmission in R2. Cumulative AI-levy receipts are shown alongside the public-service index produced by the earmarked spending rule.

5.4. Time dividend and solo enterprise

This subsection compares E4 with E1 under R2 to ask whether the AI time dividend creates additional demand through solo enterprise or mainly redirects existing spending. Figure 6 decomposes solo sales into net-new demand and displacement from incumbent firms.

E4 generates 440 solo entries, 400 exits, approximately CNY 3.165 million in sales, and a tail self-employment rate near 8%. Only 24.4% of sales are net additional demand, while 75.6% displace incumbent sales. Relative to E1, cumulative consumption falls by 3.01%, bottom-60% consumption by 3.36%, and disposable income by 2.45%. The income Atkinson index rises from 0.263 to 0.277, although liquidity vulnerability falls from 8.92% to 2.97%.

Solo enterprise therefore creates self-employment opportunities and reduces household liquidity vulnerability without necessarily expanding aggregate demand. Entry support alone can increase turnover, displace incumbent sales, and widen inequality. Converting a time dividend into broad-based growth also requires mechanisms that generate new demand rather than redistribute existing demand.



(a) Solo-enterprise net-new demand

(b) Incumbent displacement

Figure 6: Solo-enterprise demand decomposition in R2. Panel (a) reports net-new demand, while panel (b) reports solo spending displaced from incumbent firms.

5.5. Integrated compact and fiscal constraints

This subsection evaluates two nested comparisons under R2. The E5–E1 comparison tests whether an integrated policy compact can combine employment protection, redistribution, entrepreneurship, and public demand. The E6–E5 comparison asks what tighter fiscal limits add. Figure 7 links these comparisons to market concentration and cumulative expenditure curtailment.

E5 activates employment responsibility, the AI levy, solo enterprise, active transfers, procurement, and government AI. It blocks 215 AI-attributable layoffs, collects CNY 5.336 million in levy revenue, and generates CNY 3.220 million in solo sales. Relative to E1, the public-service index is 83.1% higher and liquidity vulnerability falls from 8.92% to 1.30%.

These gains are accompanied by weaker demand and firm survival. Cumulative consumption falls by 3.48%, bottom-60% consumption by 3.71%, and the ending firm count to 17. Market concentration rises to 0.112 and the income Atkinson index to 0.278. The compact therefore strengthens public services and household buffers but shifts pressure toward firm exit, concentration, and insufficient net-new demand.

The retained E6 path has a maximum debt ratio of approximately 0.31%, compared with 1.95% in E5, and 0.14% higher cumulative consumption; peak unemployment is 13.8% in both paths. The ending firm count is 16 rather than 17, and essential-cash shortfall is 19.17% rather than 17.95%. The tighter-limit path is therefore consistent with lower debt exposure but not with restored firm viability.

474 The combined result shows that an integrated compact can protect house-
475 holds and public services while transferring stress to firms, and that fiscal
476 discipline alone does not repair this imbalance. Because E5 and E6 do not
477 share frozen LLM responses, their numerical difference remains descriptive; a
478 causal fiscal comparison requires paired seeds and common response caches.

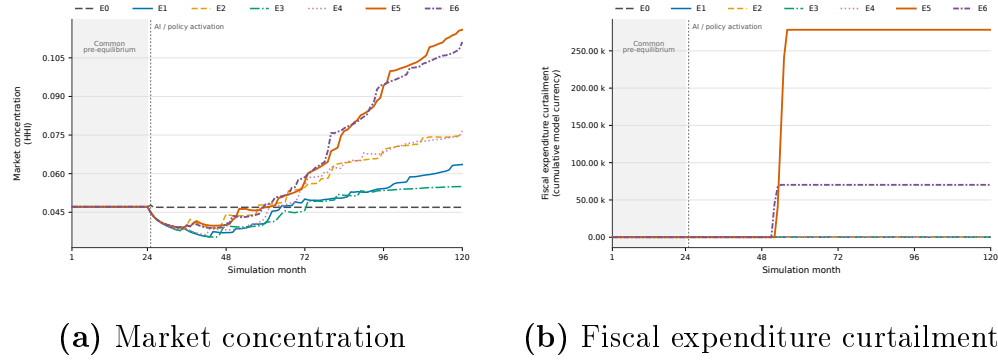


Figure 7: Market and fiscal adjustment in R2. The HHI path tracks market concentration, while cumulative expenditure curtailment shows when and how strongly fiscal limits bind.

479 5.6. LLM-participation architecture sensitivity

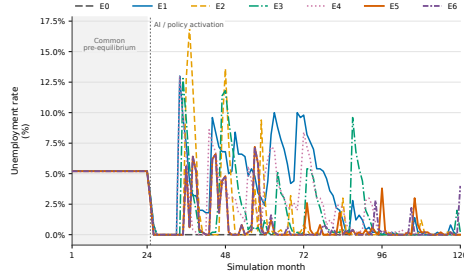
480 This subsection compares R1 and R2 across E0–E6 to distinguish sensi-
481 tivity in aggregate outcomes from sensitivity in transition dynamics. Figure 8
482 compares monthly unemployment and active-firm paths under the two LLM-
483 participation architectures.

484 Cumulative real consumption in R2 differs from R1 by only -0.48% to
485 $+0.18\%$ across the seven scenarios. The retained paths therefore show little
486 aggregate-demand sensitivity to the LLM-participation architecture.

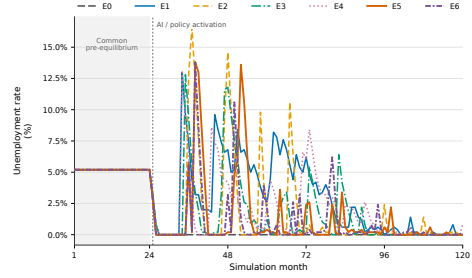
487 The adjustment paths differ substantially. Under both E5 and E6, peak
488 unemployment is 7.2% in R1 and 13.8% in R2. Ending firm counts are 20
489 versus 17 in E5 and 20 versus 16 in E6. In R2/E5, four firms exit between
490 months 34 and 40 and remove 129 jobs, concentrating the unemployment
491 peak.

492 Similar aggregate consumption therefore does not imply equivalent tran-
493 sition mechanisms. The architecture comparison shows small differences in
494 total consumption but large differences in adjustment speed, cash distress,
495 exit timing, firm survival, and peak unemployment. The path-level compar-
496 ison therefore remains necessary alongside the R2 main experiment, whose

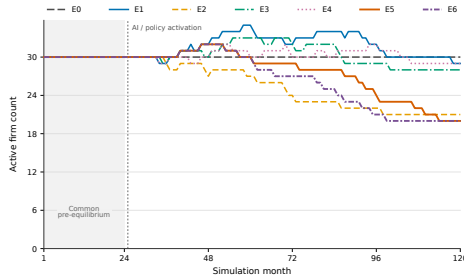
497 role was determined by the study's focus on enterprise labor adjustment and
 498 survival. Because the two architectures do not share frozen LLM responses,
 499 these differences measure combined architecture and service-response sensi-
 500 tivity rather than a fully isolated architecture effect.



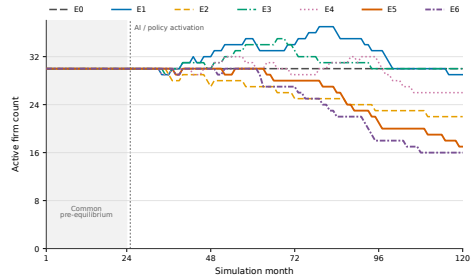
(a) R1 unemployment



(b) R2 unemployment



(c) R1 active firms



(d) R2 active firms

Figure 8: LLM-participation-sensitive transition paths across E0–E6. Monthly unemployment and active-firm counts compare R1 with R2, revealing differences in adjustment peaks and firm survival despite similar aggregate consumption.

501 6. Discussion

502 6.1. Key findings and institutional implications

503 In the current single-seed path, private AI raises cumulative consumption
504 but also produces a pronounced transition-unemployment peak. Employment
505 responsibility, AI levies, and solo enterprise all activate, yet their benefits
506 leak through firm exit, tax incidence, incumbent displacement, and market
507 concentration. E3 provides the most balanced combination of consumption,
508 employment, and public services along this path. E5 and E6 improve public
509 services and household liquidity while weakening firm survival and some con-
510 sumption outcomes. The central finding is therefore not that one scenario
511 is optimal, but that institutions redistribute transition risks, returns, and
512 binding constraints across households, firms, and government.

513 E2 converts part of the productivity dividend into lower work intensity
514 rather than immediate dismissal. If employment responsibility applies only
515 inside operating firms, however, risk can reappear through reduced hiring,
516 delayed exit, and distress exemptions. Employment policy must therefore
517 cover firm continuity as well as existing jobs by combining wage-cost sharing,
518 restructuring finance, short-time work support, training, worker protection
519 at exit, and incentives for new job creation.

520 E3 shows that part of the productivity rent can be converted into social
521 return through an auditable institutional interface. One potentially auditable
522 implementation would attach charges to measurable inputs such as compute,
523 electricity, bandwidth, or high-intensity deployment rather than attempting
524 to estimate the contribution of each model output. This is a policy-design
525 hypothesis, not an implementation result established by the simulation. The
526 current 70/30 recycling rule closes the restricted-fund identity and raises
527 public services with little aggregate-demand loss. This result is partly me-
528 chanical because the permitted uses of the fund are built into the mechanism.
529 Credible policy calibration therefore requires variation in rates, exemptions,
530 incidence, energy-efficiency rebates, and geographic leakage.

531 E4 demonstrates that solo-enterprise formation is not equivalent to net-
532 new demand. AI may reduce the fixed cost of one-person firms in consulting,
533 content, design, software, and local services, but their macroeconomic value
534 depends on external markets, new products, and business-to-business de-
535 mand rather than registration counts. Portable benefits, payment and credit
536 infrastructure, market access, and failure insurance may therefore be as im-
537 portant as entrepreneurship promotion.

538 Integrated policy is a coordination problem. E5 is not the sum of three
539 isolated effects: employment responsibility changes adjustment speed; the
540 levy changes firm costs and public spending; solo enterprise changes the
541 ownership of demand; and transfers and procurement feed back into cash
542 flow. The central design questions are who bears the cost, when collected
543 revenue is spent, and how concentrated firm exit is prevented. The retained
544 E6 path suggests that lower debt exposure need not coincide with improved
545 firm dynamics, although the unfrozen LLM responses prevent exact fiscal
546 attribution.

547 The retained LLM-participation regimes yield different transition paths.
548 Firm-level LLM use changes the action-generating architecture, while un-
549 frozen provider responses add service-level variation. Similar cumulative con-
550 sumption alongside a 6.6-percentage points difference in peak unemployment
551 shows that this combined sensitivity is an experimental dimension rather
552 than a purely engineering detail.

553 *6.2. Limitations and confirmatory research agenda*

554 The current formal matrix contains one seed and 500 household units. It
555 cannot estimate cross-seed uncertainty, sign stability, confidence intervals, or
556 scale dependence. At this population size, a small cluster of firm exits can
557 produce a substantial jump in unemployment. Differences between scenarios
558 must therefore be interpreted as path-level mechanism evidence rather than
559 a robust policy ranking. These limits concern uncertainty, sign stability, and
560 generalization; they do not by themselves invalidate within-seed common-
561 state counterfactual contrasts.

562 The no-new-AI path also contains a rapid post-fork labor adjustment:
563 unemployment falls from the 5.2% month-24 calibration anchor to 2.8% in
564 month 25 and zero in month 26. The E1–E0 contrast therefore identifies the
565 AI shock relative to an evolving no-new-AI economy, not relative to a static
566 5.2% unemployment baseline. A post-restore one-step continuity gate should
567 be added to distinguish intended delayed adjustment from a checkpoint or
568 scheduling discontinuity.

569 Behavioral qualification and architecture identification remain incom-
570 plete. Zero decision fallback establishes execution completeness, not vali-
571 dated behavioral responsiveness. The R2/E3 firm weak-demand response
572 check fails; the LLM service is not deterministic; E5/E6 responses are not
573 frozen across the fiscal contrast; and the retained architecture matrix ex-
574 cludes a complete household-LLM row. The current design therefore cannot

575 fully separate economic randomness, service randomness, and architecture
576 effects.

577 The model represents a stylized transition economy rather than a cali-
578 brated digital representation of a particular city. Marginal propensities to
579 consume, cash buffers, AI substitution, levy rates, recycling shares, solo-
580 demand decomposition, and distress exemptions remain experimental pa-
581 rameters. Interest rates, monetary policy, detailed occupation transitions,
582 endogenous training, migration, land, housing, and empirically calibrated
583 task exposure are outside the current model.

584 Confirmatory work requires at least three paired seeds and should pro-
585 gressively cover the planned 1–50 range. It should also test population scales
586 of 1,000 and 5,000, use frozen or common LLM response caches, conduct
587 pre-specified parameter sensitivities, and externally validate transition direc-
588 tions and plausible magnitudes. Until those tests are complete, the findings
589 constitute mechanism evidence and falsifiable hypotheses, not a city forecast
590 or final policy ranking.

591 *6.3. Significance for urban science*

592 AI transition is fundamentally an urban question. Employment and
593 household consumption, firm production and exit, digital infrastructure, pub-
594 lic services, and fiscal capacity are concentrated in urban systems and con-
595 nected through demand, cash flow, and institutional feedback. The con-
596 tribution of this study is therefore not limited to asking whether AI raises
597 productivity. It places productivity shocks, employment adjustment, firm
598 dynamics, household vulnerability, market structure, and public recycling
599 within one closed-loop stylized urban economy and examines how institu-
600 tions redistribute transition risk.

601 This perspective implies that urban AI policy cannot be evaluated only
602 through aggregate output or mean consumption. Improvements in macroeco-
603 nomic totals may coexist with transition-unemployment peaks, concentrated
604 firm exit, weaker bottom-group consumption, or inadequate public services.
605 Urban-science evaluation should therefore track both adjustment paths and
606 final outcomes across employment, household liquidity, firm survival, net-new
607 demand, distribution, market structure, public services, and fiscal timing.

608 AI systems also build on accumulated language, code, scientific knowl-
609 edge, public education, research infrastructure, digital infrastructure, and the
610 work of many contributors. The distribution of AI dividends cannot there-
611 fore be assessed only through ownership of models, compute, and platforms.

612 If productive capacity and distribution channels remain highly concentrated,
613 productivity gains may accrue as capital income and market power while the
614 costs of job transitions, weaker bargaining power, and reduced development
615 opportunities are dispersed across workers and urban communities.

616 The simulation cannot show that this outcome is inevitable. Within the
617 modeled mechanisms, however, productivity growth alone does not ensure
618 broadly shared benefits. Competition policy, public compute, open infras-
619 tructure, worker participation, inclusive education, lifelong learning, portable
620 social protection, and returns for knowledge and data contributions are there-
621 fore testable institutional hypotheses for converting AI gains into social re-
622 turns.

623 For urban science, the central question is not whether cities adopt AI, but
624 whether urban institutions can translate new productive capacity into sta-
625 ble employment, resilient firm dynamics, accessible public services, broadly
626 shared prosperity, and human development. Human development should re-
627 main the normative objective of technological and urban policy rather than
628 being treated as a residual outcome of technological competition. This is
629 a normative proposition and a research agenda suggested by the model’s
630 mechanisms, not an empirical conclusion already established by the present
631 simulation.

632 7. Conclusion

633 This study develops a stock–flow-aware heterogeneous-agent model of a
634 stylized urban economy to examine how institutions mediate an AI produc-
635 tivity shock. The common-state counterfactuals answer the four research
636 questions directly. First, laissez-faire private AI raises cumulative consump-
637 tion while producing a substantially larger transition-unemployment peak
638 than the no-new-AI path. Second, the three targeted institutions operate
639 through distinct mechanisms: employment responsibility converts part of la-
640 bor saving into shorter hours but does not prevent exit-related job loss; the AI
641 infrastructure levy converts a share of productivity rent into public services;
642 and solo enterprise expands self-employment but mostly reallocates existing
643 demand. Third, the integrated compact improves public services and house-
644 hold liquidity while concentrating stress in firm survival; the tighter-limit
645 E6 path has lower debt exposure without less firm-side pressure, although
646 exact attribution requires common LLM responses. Fourth, the R1–R2 com-
647 parison shows materially different unemployment and exit paths even when

648 cumulative consumption differs little; this comparison measures combined
649 architecture and service-response sensitivity.

650 The policy implication is not that one scenario is universally preferred. In-
651 stitutional design determines where transition costs appear, how quickly pro-
652 ductivity gains return to households and public services, and whether employ-
653 ment protection is accompanied by viable firms and net-new demand. The
654 model therefore supports joint evaluation of demand, employment, household
655 liquidity, distribution, firm dynamics, public services, and fiscal timing.

656 The results are mechanism-based counterfactual evidence from a styl-
657 ized system. Further paired-seed, population-scale, parameter-sensitivity,
658 response-cache, and external-validation work is required for uncertainty es-
659 timates and policy ranking. The contribution of the current study is an
660 auditable framework for turning normative claims about broadly shared AI
661 gains into explicit, falsifiable institutional mechanisms.

662 **Declaration of competing interest**

663 The authors declare that they have no known competing financial inter-
664 ests or personal relationships that could have appeared to influence the work
665 reported in this paper.

666 **Data and code availability**

667 The implementation, resolved configurations, prompt templates, decision
668 audits, retained formal results, canonical monthly dataset, data dictionary,
669 and figure-generation scripts are maintained in the project repository. The
670 current dataset contains seed 1 only. Until a public repository and archival
671 DOI are released, the documented artifacts are available from the correspond-
672 ing author on reasonable request.

673 **Declaration of generative AI and AI-assisted technologies**

674 AgentSociety coordinates bounded LLM decisions as an explicit com-
675 ponent of the research design. Separately, OpenAI Codex assisted with
676 manuscript review and language editing. No generative AI system directly
677 set monetary quantities, accounting entries, market settlement, or validation
678 outcomes. The authors reviewed all AI-assisted material and remain respon-
679 sible for the model, evidence boundaries, interpretation, and final text.

680 **Appendix A. Pre-specified institutional scenario settings**

Table A.4: Detailed pre-specified settings for the institutional scenarios (`institutional_v2`).

Scenario	Setting	Pre-specified specification
E0	Private AI	Off.
	Scenario role	Post-warm-up time counterfactual without new private AI after the common month-24 state.
	Active demand and government AI	Transfers, demand-responsive procurement, and government AI off.
	Fiscal limits	Maximum annual deficit ratio 4%; debt ratio 60%.
E1	Primary contrast	Reference state for the E1–E0 contrast.
	Private AI	On.
	Scenario role	Laissez-faire private AI operating through productivity, price, wage, and displacement channels.
	Added institution	None.
E2	Active demand and government AI	Transfers, demand-responsive procurement, and government AI off.
	Fiscal limits	Maximum annual deficit ratio 4%; debt ratio 60%.
	Primary contrast	E1–E0.
	Private AI	On.
E2	Added institution	Employment responsibility and temporary wage-cost sharing.
	Protected employment	Employment-retention floor equal to 1.00 of the no-AI labor requirement; monthly wages preserved; minimum required hours 0.
	Wage-cost sharing	Government share 0.25 for an initial six-month support window.
	Distress extension	Up to six additional months; early warnings activate after four consecutive loss months or four cash-distress months.
	Exemption and restructuring	Distress exemption allowed after eight distress months; two-month restructuring grace before otherwise eligible bankruptcy.
	Active demand and government AI	Transfers, demand-responsive procurement, and government AI off.
	Fiscal limits	Maximum annual deficit ratio 4%; debt ratio 60%.
	Primary contrast	E2–E1.

Continued on the next page.

Table A.4 continued from the previous page.

Scenario	Setting	Pre-specified specification
E3	Private AI	On.
	Added institution	AI infrastructure levy with immediate earmarked recycling.
	Levy base	AI productivity rent, bounded by positive operating surplus.
	Capture schedule	Initial capture rate 0.20, rising to 0.50 over 12 months.
	Allowance and price incidence	Basic-consumption exemption share 0.25; consumer-price pass-through 0.20.
	Same-month financing	Advance share capped at 0.80 of the conservative current-levy-revenue forecast.
	Earmarked recycling	Public services 0.70; public investment 0.30.
	Active demand and government AI	General transfers, demand-responsive procurement, and government AI off; levy-funded recycling remains active.
	Fiscal limits	Maximum annual deficit ratio 4%; debt ratio 60%.
E4	Primary contrast	E3–E1.
	Private AI	On.
	Added institution	AI time dividend and mixed-demand solo-enterprise formation.
	Entry eligibility	Incubation-readiness threshold 0.18; minimum personal-AI use rate 0.20; startup cost equal to 0.50 months of the household’s disposable-income anchor; no simultaneous wage employment and self-employment.
	Entry and capacity caps	Maximum monthly entry share 0.01; maximum self-employed share 0.10; solo-market share 0.05.
	Existing-demand channels	Substitution-market share 0.025; B2B share of firm-investment orders 0.10.
	Additional-demand channels	Induced-demand share 0.005 of baseline household demand with cash-drawdown rate 0.10; external-demand share 0.005 of baseline household demand.
	Cost and tax	Operating-cost rate 0.25; income-tax rate 0.05.
	Survival and AI use	Minimum income ratio 0.35; failure window six months; self-employed AI target 0.80.
	Active demand and government AI	Transfers, demand-responsive procurement, and government AI off.
	Fiscal limits	Maximum annual deficit ratio 4%; debt ratio 60%.
	Primary contrast	E4–E1.

Continued on the next page.

Table A.4 continued from the previous page.

Scenario	Setting	Pre-specified specification
E5	Private AI	On.
	Institutional portfolio	All pre-specified E2 employment-responsibility, E3 levy/recycling, and E4 solo-enterprise mechanisms operate jointly with the parameter values above.
	Active demand and government AI	Transfers, demand-responsive procurement, and government AI on; active-demand policy strategy.
	Active-demand parameters	Procurement-response multiplier 1.75; transfer-response multiplier 0.80; employment-support rate 0.12.
E6	Fiscal limits	Maximum annual deficit ratio 4%; debt ratio 60%.
	Primary contrast	E5–E1.
	Private AI	On.
	Institutional portfolio	Same pre-specified institutional portfolio and active-demand parameters as E5.
	Active demand and government AI	Transfers, demand-responsive procurement, and government AI on; active-demand policy strategy.
	Fiscal limits	Maximum annual deficit ratio 2%; debt ratio 40%.
	Primary contrast	E6–E5.

681 *Notes:* The table reports the primary parameter values fixed before the re-
682 tained runs. They are stored in the following repository file:

683 `ai_economy_execution/config/baseline.json`

684 Fiscal limits are the maximum annual deficit ratio and debt ratio. All
685 branches share the qualified month-24 state and activate in month 25. The
686 settings are exploratory counterfactuals rather than empirical estimates for
687 a specific city.

688 **Appendix B. Supplementary provenance and claim boundary**

689 The canonical research dataset contains 1,680 monthly rows and 473
690 columns. All 14 R1/R2 by E0–E6 paths are complete, no decision required
691 a fallback, and the accounting, boundary, warm-up, and common-path gates
692 pass. The full shared source fingerprint, beginning `aa6ab84ed880`, is retained
693 in the run artifacts. Behavioral monotonicity coverage remains incomplete,
694 and R2/E3 fails the firm weak-demand response direction check. The arti-
695 facts therefore support mechanism activation and path-level leakage claims,
696 but not multi-seed uncertainty estimates, robust policy rankings, validated
697 LLM behavioral responsiveness, or predictions for a specific city.

698 References

- 699 Acemoglu, D., 2025. The simple macroeconomics of AI. *Economic Policy* 40,
700 13–58. doi:[10.1093/epolic/eiae042](https://doi.org/10.1093/epolic/eiae042).
- 701 Acemoglu, D., Restrepo, P., 2018. The race between man and machine: Im-
702 plications of technology for growth, factor shares, and employment. *Ameri-*
703 *can Economic Review* 108, 1488–1542. doi:[10.1257/aer.20160696](https://doi.org/10.1257/aer.20160696).
- 704 Acemoglu, D., Restrepo, P., 2020. Robots and jobs: Evidence from US labor
705 markets. *Journal of Political Economy* 128, 2188–2244. doi:[10.1086/](https://doi.org/10.1086/705716)
706 [705716](https://doi.org/10.1086/705716).
- 707 Argyle, L.P., Busby, E.C., Fulda, N., Gubler, J.R., Rytting, C., Wingate,
708 D., 2023. Out of one, many: Using language models to simulate human
709 samples. *Political Analysis* 31, 337–351. doi:[10.1017/pan.2023.2](https://doi.org/10.1017/pan.2023.2).
- 710 Atkinson, A.B., 1970. On the measurement of inequality. *Journal of Eco-*
711 *nomic Theory* 2, 244–263. doi:[10.1016/0022-0531\(70\)90039-6](https://doi.org/10.1016/0022-0531(70)90039-6).
- 712 Auerbach, A.J., Feenberg, D., 2000. The significance of federal taxes as
713 automatic stabilizers. *Journal of Economic Perspectives* 14, 37–56. doi:[10.](https://doi.org/10.1257/jep.14.3.37)
714 [1257/jep.14.3.37](https://doi.org/10.1257/jep.14.3.37).
- 715 Autor, D.H., Levy, F., Murnane, R.J., 2003. The skill content of recent
716 technological change: An empirical exploration. *The Quarterly Journal of*
717 *Economics* 118, 1279–1333. doi:[10.1162/003355303322552801](https://doi.org/10.1162/003355303322552801).
- 718 Bail, C.A., 2024. Can generative AI improve social science? *Proceedings of*
719 *the National Academy of Sciences* 121, e2314021121. doi:[10.1073/pnas.](https://doi.org/10.1073/pnas.2314021121)
720 [2314021121](https://doi.org/10.1073/pnas.2314021121).
- 721 Berg, A., Buffie, E.F., Zanna, L.F., 2018. Should we fear the robot revo-
722 lution? (the correct answer is yes). *Journal of Monetary Economics* 97,
723 117–148. doi:[10.1016/j.jmoneco.2018.05.014](https://doi.org/10.1016/j.jmoneco.2018.05.014).
- 724 Bessen, J.E., 2019. AI and jobs: The role of demand, in: Agrawal, A.,
725 Gans, J., Goldfarb, A. (Eds.), *The Economics of Artificial Intelligence:*
726 *An Agenda*. University of Chicago Press, Chicago, pp. 291–307. URL:
727 <https://www.nber.org/papers/w24235>. earlier version: NBER Working
728 Paper 24235.

- 729 Brynjolfsson, E., Li, D., Raymond, L.R., 2025. Generative AI at work. The
730 Quarterly Journal of Economics 140, 889–942. doi:[10.1093/qje/qjae044](https://doi.org/10.1093/qje/qjae044).
- 731 Caiani, A., Godin, A., Caverzasi, E., Gallegati, M., Kinsella, S., Stiglitz,
732 J.E., 2016. Agent based-stock flow consistent macroeconomics: Towards a
733 benchmark model. Journal of Economic Dynamics and Control 69, 375–
734 408. doi:[10.1016/j.jedc.2016.06.001](https://doi.org/10.1016/j.jedc.2016.06.001).
- 735 Carroll, C.D., 1997. Buffer-stock saving and the life cycle/permanent income
736 hypothesis. The Quarterly Journal of Economics 112, 1–55. doi:[10.1162/003355397555109](https://doi.org/10.1162/003355397555109).
- 737
- 738 Dawid, H., Delli Gatti, D., 2018. Agent-based macroeconomics, in: Hommes,
739 C., LeBaron, B. (Eds.), Handbook of Computational Economics. Elsevier.
740 volume 4, pp. 63–156. doi:[10.1016/bs.hescom.2018.02.006](https://doi.org/10.1016/bs.hescom.2018.02.006).
- 741 Deaton, A., 1991. Saving and liquidity constraints. Econometrica 59, 1221–
742 1248. doi:[10.2307/2938366](https://doi.org/10.2307/2938366).
- 743 Dosi, G., Fagiolo, G., Napoletano, M., Roventini, A., Treibich, T., 2013.
744 Income distribution, credit and fiscal policies in an agent-based Keynesian
745 model. Journal of Economic Dynamics and Control 37, 1598–1625. doi:[10.1016/j.jedc.2012.11.008](https://doi.org/10.1016/j.jedc.2012.11.008).
- 746
- 747 Dosi, G., Fagiolo, G., Roventini, A., 2010. Schumpeter meeting keynes: A
748 policy-friendly model of endogenous growth and business cycles. Journal
749 of Economic Dynamics and Control 34, 1748–1767. doi:[10.1016/j.jedc.2010.06.018](https://doi.org/10.1016/j.jedc.2010.06.018).
- 750
- 751 Dosi, G., Pereira, M.C., Roventini, A., Virgillito, M.E., 2022. Technological
752 paradigms, labour creation and destruction in a multi-sector agent-based
753 model. Research Policy 51, 104565. doi:[10.1016/j.respol.2022.104565](https://doi.org/10.1016/j.respol.2022.104565).
- 754 Driess, D., Xia, F., Sajjadi, M.S.M., Lynch, C., Chowdhery, A., et al., 2023.
755 PaLM-E: An embodied multimodal language model, in: Proceedings of
756 the 40th International Conference on Machine Learning, pp. 8469–8488.
757 URL: <https://proceedings.mlr.press/v202/driess23a.html>.
- 758 Eloundou, T., Manning, S., Mishkin, P., Rock, D., 2024. GPTs are GPTs:
759 Labor market impact potential of LLMs. Science 384, 1306–1308. doi:[10.1126/science.adj0998](https://doi.org/10.1126/science.adj0998).
- 760

- 761 Friedman, M., 1957. A Theory of the Consumption Function. Princeton
762 University Press, Princeton, NJ.
- 763 Gao, C., et al., 2024. Large language models empowered agent-based mod-
764 eling and simulation: A survey and perspectives. Humanities and Social
765 Sciences Communications 11, 1259. doi:[10.1057/s41599-024-03611-3](https://doi.org/10.1057/s41599-024-03611-3).
- 766 Grimm, V., Berger, U., Bastiansen, F., Eliassen, S., Ginot, V., Giske, J.,
767 et al., 2006. A standard protocol for describing individual-based and
768 agent-based models. Ecological Modelling 198, 115–126. doi:[10.1016/
769 j.ecolmodel.2006.04.023](https://doi.org/10.1016/j.ecolmodel.2006.04.023).
- 770 Heathcote, J., 2005. Fiscal policy with heterogeneous agents and incomplete
771 markets. The Review of Economic Studies 72, 161–188. doi:[10.1111/
772 0034-6527.00328](https://doi.org/10.1111/0034-6527.00328).
- 773 Kaplan, G., Moll, B., Violante, G.L., 2018. Monetary policy according to
774 HANK. American Economic Review 108, 697–743. doi:[10.1257/aer.
775 20160042](https://doi.org/10.1257/aer.20160042).
- 776 Keynes, J.M., 1936. The General Theory of Employment, Interest and
777 Money. Macmillan, London.
- 778 Kleijnen, J.P.C., 1988. Analyzing simulation experiments with common ran-
779 dom numbers. Management Science 34, 65–74. doi:[10.1287/mnsc.34.1.
780 65](https://doi.org/10.1287/mnsc.34.1.65).
- 781 Li, Y., et al., 2022. Competition-level code generation with AlphaCode.
782 Science 378, 1092–1097. doi:[10.1126/science.abq1158](https://doi.org/10.1126/science.abq1158).
- 783 McKay, A., Reis, R., 2016. The role of automatic stabilizers in the U.S.
784 business cycle. Econometrica 84, 141–194. doi:[10.3982/ECTA11574](https://doi.org/10.3982/ECTA11574).
- 785 National Bureau of Statistics of China, 2025a. Household income and con-
786 sumption expenditure in 2024. Official statistical release. URL: [https:
787 //www.stats.gov.cn/sj/zxfb/202501/t20250117_1958325.html](https://www.stats.gov.cn/sj/zxfb/202501/t20250117_1958325.html).
- 788 National Bureau of Statistics of China, 2025b. Statistical communiqué of
789 the people’s republic of china on the 2024 national economic and social
790 development. Official statistical communiqué. URL: [https://www.stats.
791 gov.cn/sj/zxfb/202502/t20250228_1958817.html](https://www.stats.gov.cn/sj/zxfb/202502/t20250228_1958817.html).

792 National Bureau of Statistics of China, 2026a. Household income and con-
793 sumption expenditure in 2025. Official statistical release. URL: https://www.stats.gov.cn/sj/zxfb/202601/t20260119_1962321.html.
794

795 National Bureau of Statistics of China, 2026b. Statistical communiqué of
796 the people’s republic of china on the 2025 national economic and social
797 development. Official statistical communiqué. URL: https://www.stats.gov.cn/sj/zxfb/202602/t20260228_1962662.html.
798

799 National Financial Regulatory Administration, 2023. Measures for
800 the capital management of commercial banks. Official regula-
801 tion. URL: <https://www.nfra.gov.cn/cn/view/pages/rulesDetail.html?docId=1134197&itemId=4214>.
802

803 National Financial Regulatory Administration, 2026. Banking and insur-
804 ance sector performance in the fourth quarter of 2025. Official statistical
805 release. URL: <https://www.nfra.gov.cn/cn/view/pages/ItemDetail.html?docId=1249435&itemId=915>.
806

807 Noy, S., Zhang, W., 2023. Experimental evidence on the productivity effects
808 of generative artificial intelligence. *Science* 381, 187–192. doi:[10.1126/science.adh2586](https://doi.org/10.1126/science.adh2586).
809

810 OpenAI, 2023. GPT-4 technical report. doi:[10.48550/arXiv.2303.08774](https://doi.org/10.48550/arXiv.2303.08774),
811 [arXiv:2303.08774](https://arxiv.org/abs/2303.08774).

812 Pan, C., Yaman, B., Nesti, T., Mallik, A., Allievi, A.G., Velipasalar, S., Ren,
813 L., 2024. VLP: Vision language planning for autonomous driving, in: Pro-
814 ceedings of the IEEE/CVF Conference on Computer Vision and Pattern
815 Recognition, pp. 14760–14769. doi:[10.1109/CVPR52733.2024.01398](https://doi.org/10.1109/CVPR52733.2024.01398).

816 Park, J.S., O’Brien, J., Cai, C.J., Morris, M.R., Liang, P., Bernstein, M.S.,
817 2023. Generative agents: Interactive simulacra of human behavior, in: Pro-
818 ceedings of the 36th Annual ACM Symposium on User Interface Software
819 and Technology, pp. 1–22. doi:[10.1145/3586183.3606763](https://doi.org/10.1145/3586183.3606763).

820 People’s Bank of China, 2025. China monetary policy report,
821 quarter one 2025. Official monetary policy report. URL:
822 [https://www.pbc.gov.cn/zhengcehuobisi/125207/125227/125957/
823 f5c4690f2cbd40918bf24c2d39ac58af/2025091218344076234/
824 2025081517321679368.pdf](https://www.pbc.gov.cn/zhengcehuobisi/125207/125227/125957/f5c4690f2cbd40918bf24c2d39ac58af/2025091218344076234/2025081517321679368.pdf).

- 825 Piao, J., Yan, Y., Zhang, J., Li, N., Yan, J., Lan, X., et al., 2025.
 826 AgentSociety: Large-scale simulation of LLM-driven generative agents
 827 advances understanding of human behaviors and society. URL: <https://arxiv.org/abs/2502.08691>, [arXiv:2502.08691](https://arxiv.org/abs/2502.08691).
 828
- 829 Piao, J., Zhang, J., Huang, H., Zhang, K., Wang, J.Y., Zhao, X., Li, S.,
 830 Sun, B., Chang, J., Xu, F., Wang, C., et al., 2026. AgentSociety 2:
 831 An integrated research environment for executable social science. URL:
 832 <https://arxiv.org/abs/2607.11895>, [arXiv:2607.11895](https://arxiv.org/abs/2607.11895).
- 833 Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., Chen, M., 2022. Hierarchi-
 834 cal text-conditional image generation with CLIP latents. doi:[10.48550/](https://arxiv.org/abs/2204.06125)
 835 [arXiv.2204.06125](https://arxiv.org/abs/2204.06125), [arXiv:2204.06125](https://arxiv.org/abs/2204.06125).
- 836 Schumpeter, J.A., 1942. Capitalism, Socialism and Democracy. Harper &
 837 Brothers, New York.
- 838 Solow, R.M., 1957. Technical change and the aggregate production func-
 839 tion. The Review of Economics and Statistics 39, 312–320. doi:[10.2307/](https://arxiv.org/abs/1926047)
 840 [1926047](https://arxiv.org/abs/1926047).
- 841 State Council of the People’s Republic of China, 2025. Government work
 842 report 2025. Official government report. URL: [https://www.gov.cn/](https://www.gov.cn/yaowen/liebiao/202503/content_7010168.htm)
 843 [yaowen/liebiao/202503/content_7010168.htm](https://www.gov.cn/yaowen/liebiao/202503/content_7010168.htm).